Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2020 – Paper 3 – Solutions**

June 2020 – Question 1 – Solution

(a) Prove by mathematical induction that $\sum_{t=1}^{n}(2t-1) = n^2, n \in \mathbb{N}$. [8]

Let $P(n)$ be the statement:

$\sum_{t=1}^{n}(2t-1) = n^2, n \in \mathbb{N}$

Looking at $P(1)$.

L.H.S. $= 2(1) - 1$

$= 2 - 1$

$= 1$

R.H.S. $= (1)^2$

$= 1$

Since L.H.S. = R.H.S, then $P(1)$ is true.

$\therefore P(n)$ is true for $n = 1$.

Assume $P(k)$ is true.

That is, $\sum_{t=1}^{k}(2t-1) = k^2$

Looking at $P(k+1)$.

That is, $\sum_{t=1}^{k+1}(2t-1) = (k+1)^2$

We need to prove $P(k+1)$.

$$\sum_{t=1}^{k+1}(2t-1) = \sum_{t=1}^{k}(2t-1) + (k+1)\text{th term}$$

$$= k^2 + 2(k+1) - 1 \qquad \text{[by assumption]}$$

$$= k^2 + 2k + 2 - 1$$

$$= k^2 + 2k + 1$$

$$= (k+1)^2$$

$$= \text{R.H.S.}$$

$\therefore$ If $P(k)$ is true, $P(k+1)$ is also true.

Since $P(1)$ is true, and since $P(k+1)$ is true when $P(k)$ is true, then by the principle of mathematical induction, $P(n)$ is true for all n, ie.

$$\sum_{t=1}^{n}(2t-1) = n^2, n \in \mathbb{N}$$

Q.E.D.

(b) Solve the equation $3(2^{x+4}) = 360$. [4]

$3(2^{x+4}) = 360$

$2^{x+4} = \frac{360}{3}$

$2^{x+4} = 120$

Taking ln on both sides:

$\ln 2^{x+4} = \ln 120$

$(x+4)\ln 2 = \ln 120$

$x + 4 = \frac{\ln 120}{\ln 2}$

$x = \frac{\ln 120}{\ln 2} - 4$

$x = 2.91$ (to 3 significant figures)

$\therefore$ The solution is $x = 2.91$.

(c) Using the Remainder Theorem, or otherwise, show that

$$x^6 - b^6 = (x+b)(x-b)(x^2 - bx + b^2)(x^2 + bx + b^2) \qquad [8]$$

Taking L.H.S:

$$x^6 - b^6 = (x^2)^3 - (b^2)^3$$

$$= (x^2 - b^2)(x^4 + b^2x^2 + b^4)$$

$$= (x+b)(x-b)(x^4 + b^2x^2 + b^4) \qquad \rightarrow \text{Equation 1}$$

Now,

$$x^4 + b^2x^2 + b^4 = x^4 + 2b^2x^2 + b^4 - b^2x^2$$

$$= (x^2 + b^2)^2 - (bx)^2$$

$$= (x^2 + b^2 - bx)(x^2 + b^2 + bx) \qquad \text{[difference of two squares]}$$

$$= (x^2 - bx + b^2)(x^2 + bx + b^2) \qquad \rightarrow \text{Equation 2}$$

Substituting Equation 2 into Equation 1 gives:

$$x^6 - b^6 = (x+b)(x-b)(x^2 - bx + b^2)(x^2 + bx + b^2)$$

$$= \text{R.H.S.}$$

$$\therefore x^6 - b^6 = (x+b)(x-b)(x^2 - bx + b^2)(x^2 + bx + b^2)$$

Q.E.D.

Total: 20 marks

June 2020 – Question 2 – Solution

(a) Prove that $\frac{\sin\theta}{1+\cos\theta}+\frac{1+\cos\theta}{\sin\theta}=\frac{2}{\sin\theta}$. [5]

Taking L.H.S:

$$\frac{\sin\theta}{1+\cos\theta}+\frac{1+\cos\theta}{\sin\theta}=\frac{\sin^2\theta+(1+\cos\theta)^2}{\sin\theta(1+\cos\theta)}$$

$$=\frac{\sin^2\theta+1+2\cos\theta+\cos^2\theta}{\sin\theta(1+\cos\theta)}$$

$$=\frac{(\sin^2\theta+\cos^2\theta)+1+2\cos\theta}{\sin\theta(1+\cos\theta)}$$

$$=\frac{1+1+2\cos\theta}{\sin\theta(1+\cos\theta)}$$

$$=\frac{2+2\cos\theta}{\sin\theta(1+\cos\theta)}$$

$$=\frac{2(1+\cos\theta)}{\sin\theta(1+\cos\theta)}$$

$$=\frac{2}{\sin\theta}$$

$$=\text{R.H.S.}$$

$$\therefore \frac{\sin\theta}{1+\cos\theta}+\frac{1+\cos\theta}{\sin\theta}=\frac{2}{\sin\theta}$$

Q.E.D.

(b) A curve is defined parametrically as $x = 3\sin\theta$, $y = 4\cos\theta$.

(i) Show that the Cartesian equation of the curve is $\frac{x^2}{9} + \frac{y^2}{16} = 1$. [3]

$x = 3\sin\theta$ and $y = 4\cos\theta$

$\frac{x}{3} = \sin\theta$ $\frac{y}{4} = 4\cos\theta$

Now,

$\sin^2\theta + \cos^2\theta = 1$

$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{4}\right)^2 = 1$

$\frac{x^2}{9} + \frac{y^2}{16} = 1$

$\therefore$ The Cartesian equation of the curve is $\frac{x^2}{9} + \frac{y^2}{16} = 1$.

Q.E.D.

(ii) Show that the curve and the line $y = x$ intersect at $x = \frac{12}{5}$ and $x = -\frac{12}{5}$. [2]

$\frac{x^2}{9} + \frac{y^2}{16} = 1$ $\rightarrow$ Equation 1

$y = x$ $\rightarrow$ Equation 2

Substituting Equation 2 into Equation 1 gives:

$\frac{x^2}{9} + \frac{x^2}{16} = 1$

$(\times 144)$

$$16x^2 + 9x^2 = 144$$

$$25x^2 = 144$$

$$x^2 = \frac{144}{25}$$

$$x = \pm\sqrt{\frac{144}{25}}$$

$$x = \pm\frac{12}{5}$$

Either $x = -\frac{12}{5}$ or $x = \frac{12}{5}$.

$\therefore$ The curve and the line $y = x$ intersect at $x = \frac{12}{5}$ and$x = -\frac{12}{5}$.

Q.E.D.

(iii) Given that the gradient of the curve is $\frac{-16x}{9y}$, determine the equation of ONE tangent to the curve at a point where it intersects the line $y = x$. [4]

The equation is $y = x$.

When $x = \frac{12}{5}$,

$y = \frac{12}{5}$

The point is $\left(\frac{12}{5}, \frac{12}{5}\right)$.

Now,

$\frac{dy}{dx} = \frac{-16x}{9y}$

Substituting point $\left(\frac{12}{5}, \frac{12}{5}\right)$ gives:

$$\frac{dy}{dx} = \frac{-16\left(\frac{12}{5}\right)}{9\left(\frac{12}{5}\right)}$$

$$= -\frac{16}{9}$$

Gradient of tangent $= -\frac{16}{9}$

Substituting $m = -\frac{16}{9}$ and point $\left(\frac{12}{5}, \frac{12}{5}\right)$ into $y - y_1 = m(x - x_1)$ gives:

$$y - \frac{12}{5} = -\frac{16}{9}\left(x - \frac{12}{5}\right)$$

$$y - \frac{12}{5} = -\frac{16}{9}x + \frac{64}{15}$$

$$y = -\frac{16}{9}x + \frac{64}{15} + \frac{12}{5}$$

$$y = -\frac{16}{9}x + \frac{20}{3}$$

$(\times 9)$

$$9y = -16x + 60$$

$\therefore$ The equation of one tangent to the curve is: $9y = -16x + 60$

(c) Solve $\sec^2 x - 2\tan x = 4$ for values $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. [6]

$$\sec^2 x - 2\tan x = 4$$

$$(1 + \tan^2 x) - 2\tan x = 4$$

$$\tan^2 x - 2\tan x + 1 - 4 = 0$$

$$\tan^2 x - 2\tan x - 3 = 0$$

$$(\tan x + 1)(\tan x - 3) = 0$$

Either $\tan x + 1 = 0$ or $\tan x - 3 = 0$

$\tan x = -1$ $\tan x = 3$

Consider,

$\tan x = -1$

$x = \tan^{-1}(-1)$

$x = -\frac{\pi}{4}$

Consider,

$\tan x = 3$

$x = \tan^{-1}(3)$

$x = 1.25^c$ (to 3 significant figures)

$\therefore x = -\frac{\pi}{4}, 1.25^c$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.

Total: 20 marks

June 2020 – Question 3 – Solution

(a) Water is dripping from a tap into a cylindrical bowl of radius, $r = 10\ cm$. The depth, $h\ cm$, of the water in the bowl at time, t seconds, is given by $h = \sqrt{1+4t}$.

Determine the rate of change of the volume of water in the bowl at $t = 2$ seconds. [4]

$V = \pi r^2 h$

When $r = 10$,

$V = \pi(10)^2 h$

$V = 100\pi h$

$\frac{dV}{dh} = 100\pi$

$h = \sqrt{1+4t}$

$h = (1+4t)^{\frac{1}{2}}$

$\frac{dh}{dt} = \frac{1}{2}(4)(1+4t)^{\frac{1}{2}-1}$

$\frac{dh}{dt} = 2(1+4t)^{-\frac{1}{2}}$

When $t = 2$,

$\frac{dh}{dt} = 2[1+4(2)]^{-\frac{1}{2}}$

$= 2(1+8)^{-\frac{1}{2}}$

$= 2(9)^{-\frac{1}{2}}$

$= 2\left(\frac{1}{3}\right)$

$= \frac{2}{3}$

Using the chain rule,

$$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$$

$$= 100\pi \times \frac{2}{3}$$

$$= \frac{200\pi}{3}\ cm^3 s^{-1}$$

$\therefore$ The rate of change of the volume of water in the bowl at $t = 2$ seconds is

$\frac{200\pi}{3}\ cm^3 s^{-1}$.

(b) Differentiate $f(x) = \sqrt{x^2 + 1}$ from first principles. [7]

$f(x) = \sqrt{x^2 + 1}$

Using first principles,

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{\sqrt{(x+h)^2+1} - \sqrt{x^2+1}}{h}$$

$$= \lim_{h \to 0} \frac{\sqrt{(x+h)^2+1} - \sqrt{x^2+1}}{h} \cdot \frac{\sqrt{(x+h)^2+1} + \sqrt{x^2+1}}{\sqrt{(x+h)^2+1} + \sqrt{x^2+1}}$$

$$= \lim_{h \to 0} \frac{(x+h)^2+1-(x^2+1)}{h\left(\sqrt{(x+h)^2+1} + \sqrt{x^2+1}\right)}$$

$$= \lim_{h \to 0} \frac{x^2+2xh+h^2+1-x^2-1}{h\left(\sqrt{(x+h)^2+1} + \sqrt{x^2+1}\right)}$$

$$= \lim_{h \to 0} \frac{2xh+h^2}{h\left(\sqrt{(x+h)^2+1} + \sqrt{x^2+1}\right)}$$

$$= \lim_{h \to 0} \frac{h(2x+h)}{h\left(\sqrt{(x+h)^2+1} + \sqrt{x^2+1}\right)}$$

$$= \lim_{h \to 0} \frac{2x+h}{\sqrt{(x+h)^2+1} + \sqrt{x^2+1}}$$

$$= \frac{2x+(0)}{\sqrt{(x+0)^2+1} + \sqrt{x^2+1}}$$

$$= \frac{2x}{\sqrt{x^2+1} + \sqrt{x^2+1}}$$

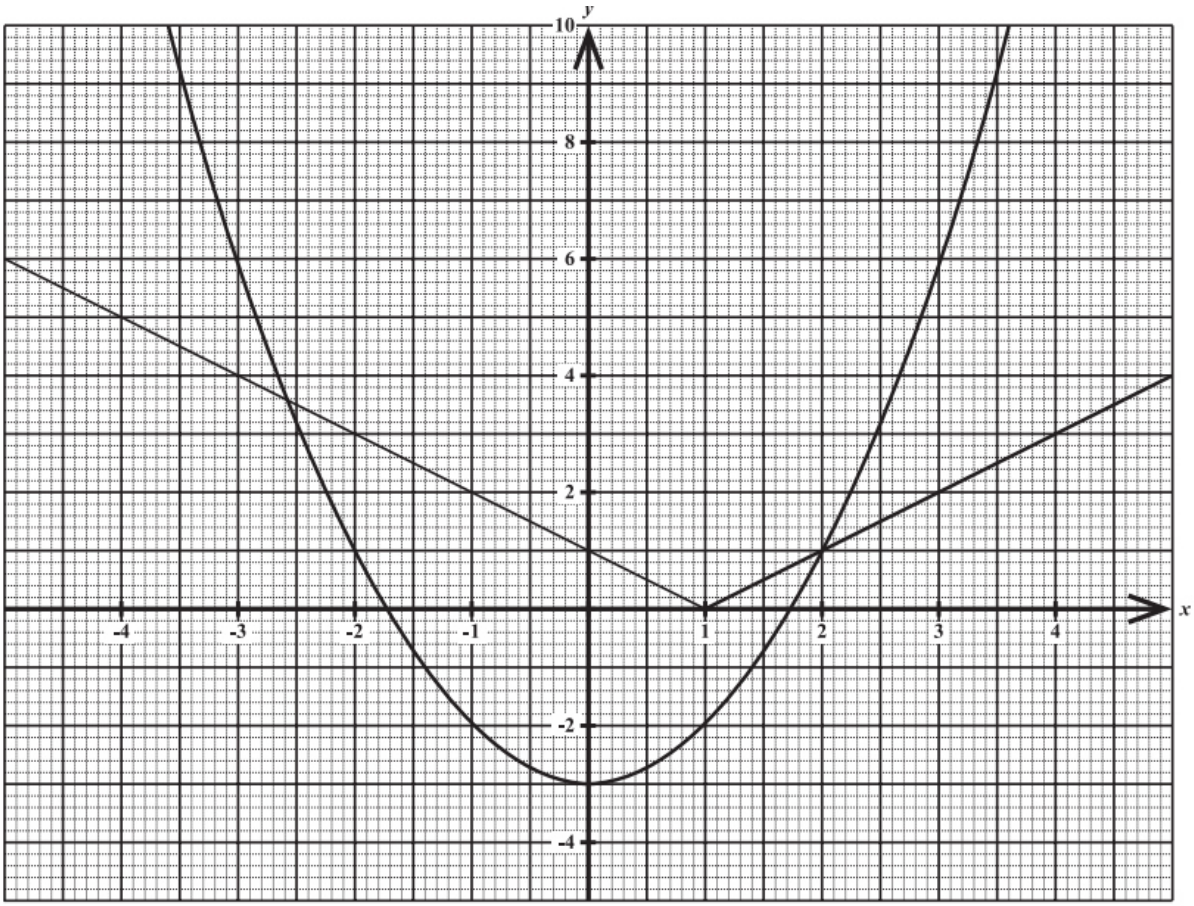
$$= \frac{2x}{2\sqrt{x^2+1}}$$

$$= \frac{x}{\sqrt{x^2+1}}$$

$$\therefore f'(x) = \frac{x}{\sqrt{x^2+1}}$$

(c) The diagram below shows the graphs of $y = |x - 1|$ and $y = x^2 - 3$.

Calculate the area enclosed by the graphs. [9]



Let us split the area into two parts, A_1 and A_2.

A_1 is between $x = -2.6$ and $x = 1$.

The line is:

$y_1 = -(x - 1)$

$= -x + 1$

$= 1 - x$

The curve is:

$y_2 = x^2 - 3$

Area of A_1 = Area under line − Area under curve

$$= \int_a^b y_1\,dx - \int_a^b y_2\,dx$$

$$= \int_{-2.6}^{1}(1-x)\,dx - \int_{-2.6}^{1}(x^2-3)\,dx$$

$$= \int_{-2.6}^{1}(1-x-x^2+3)\,dx$$

$$= \int_{-2.6}^{1}(4-x-x^2)\,dx$$

$$= \left[4x - \frac{x^2}{2} - \frac{x^3}{3}\right]_{-2.6}^{1}$$

$$= \left[4(-2.6) - \frac{(-2.6)^2}{2} - \frac{(-2.6)^3}{3}\right] - \left[4(1) - \frac{(1)^2}{2} - \frac{(1)^3}{3}\right]$$

$$= \left(-10.4 - 3.38 + \frac{2197}{375}\right) - \left(4 - \frac{1}{2} - \frac{1}{3}\right)$$

$$= \left(-\frac{5941}{750}\right) - \frac{19}{6}$$

$$= 11.088 \text{ units}^2$$

Now,

A_2 is between $x = 1$ and $x = 2$.

The line is: $y_1 = x - 1$

The curve is: $y_2 = x^2 - 3$

Area of A_2 = Area under line − Area under curve

$$= \int_a^b y_1\,dx - \int_a^b y_2\,dx$$

$$= \int_1^2 (x-1)\,dx - \int_1^2 (x^2-3)\,dx$$

$$= \int_1^2 (x-1-x^2+3)\,dx$$

$$= \int_1^2 (x-x^2+2)\,dx$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{3} + 2x\right]_1^2$$

$$= \left[\frac{(2)^2}{2} - \frac{(2)^3}{3} + 2(2)\right] - \left[\frac{(1)^2}{2} - \frac{(1)^3}{3} + 2(1)\right]$$

$$= \left(2 - \frac{8}{3} + 4\right) - \left(\frac{1}{2} - \frac{1}{3} + 2\right)$$

$$= \frac{10}{3} - \frac{13}{6}$$

$$= \frac{7}{6} \text{ units}^2$$

Hence,

Total area $= A_1 + A_2$

$$= 11.088 + \frac{7}{6}$$

$$= 12.3 \text{ units}^2 \qquad \text{(to 3 significant figures)}$$

$\therefore$ The area enclosed by the graphs is 12.3 units^2.

Total: 20 marks